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Short Communication

Correlation between lipid profile and cognitive function deterioration in ischemic stroke patients

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ABSTRACT

Context: The second most common type of dementia is vascular dementia, caused by cerebrovascular disease. The decline in cognitive function after ischemic stroke occurs with varying degrees of severity and onset. This variability is attributed to the overlapping neurodegenerative and vascular damage.

Aims: To investigate the correlation between lipid profile, a biomarker of vascular dysfunction, and cognitive function in ischemic stroke patients.

Methods: A total of 52 respondents (mean age 64.21 ± 9.78 years) were recruited, and the Spearman correlation test found a moderate positive correlation (r=0.658, p=0.04) between lipid profile and cognitive function deterioration. The ordinal regression analysis was used to calculate the odds ratio (OR).

Results: The results showed that the OR for cognitive function deterioration associated with high-density lipoprotein (HDL), low-density lipoprotein (LDL), triglycerides, and total cholesterol were -1.54 (95% CI -0.98 - -3.46, p=0.002), 1.78 (95% CI 1.27-2.46, p=0.000), 2.48 (95% CI 1.39-4.92, p=0.04), and 1.54 (95% CI 1.04-2.48, p=0.003) respectively.

Conclusion: Based on the results, a positive correlation with moderate strength was observed between lipid profile and cognitive function deterioration in ischemic stroke patients.

Keywords: cognitive function deterioration; ischemic stroke; lipid profile.

INTRODUCTION

Stroke is a disease caused by cerebrovascular disorders that persist for more than 24 hours. There are two primary types of strokes, namely ischemic, which occurs when a blood vessel is blocked, and hemorrhagic when a blood vessel bursts. In both cases, these conditions result in the deprivation of nutrients and oxygen to the brain parenchyma, leading to cell death (Kusniowati, 2013). According to the Indonesian Ministry of Health, stroke prevalence increased from 7 per 1,000 population in 2013 to 10.9 per 1,000 population in 2018 (Kesehatan Kementerian, 2018), with the majority being ischemic stroke (Dinata et al., 2013).

Stroke is the highest etiology of disability worldwide. Disability due to this condition causes a loss of productivity as well as an economic and social burden on the family (Misbach and Wendra, 2000). Risk factors for stroke include hypertension, age over 50 years, increased total cholesterol, and high blood sugar. Dinata et al. (2013) found that in Central Java, 69.79% of stroke patients had dyslipidemia.

An increase in total cholesterol levels potentially causes atherosclerosis, leading to blood vessel blockage. Furthermore, the risk of vascular dementia is increased in patients with elevated serum total cholesterol level (Pappolla et al., 2003). The apolipoprotein E (apoE) genotype reportedly plays a significant role in amyloid-plaque formation. The risk of developing Alzheimer's dementia is greatly influenced by gender, age, and lipid profile, particularly the total cholesterol level (Jarvik et al., 1995). A study found significant differences in the serum cholesterol levels between individuals with Alzheimer's disease and healthy group (Popp et al., 2013). In a cohort study on dementia patients, cholesterol levels were significantly greater at the first measurement but did not differ statistically in the second, performed after 11 to 26 years (Kivipelto et al., 2001). This was attributed to altered lipid metabolism in the geriatric population, which showed significantly increased cholesterol levels among individuals in this age group (Zhou et al., 2020). Meanwhile, in another study, serum total cholesterol levels were not significantly different between subjects with vascular dementia patients and healthy controls, but high-density lipoprotein (HDL) serum levels were lower, and serum low-density lipoprotein (LDL) levels were higher (Genin et al., 2011).

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The role of lipid profiles in post-stroke cognitive decline is currently controversial. Overlapping between neurodegenerative and vascular mechanisms is suspected to be a contributing factor. Therefore, this study aimed to investigate correlation between lipid profile, a biomarker of vascular dysfunction, and cognitive function in ischemic stroke patients.

METHODOLOGY

This cross-sectional observational analytic study aimed to determine the correlation between lipid profile and cognitive function deterioration. It was conducted at Dr. Moewardi General Hospital, Indonesia, from January to September 2022 with the consecutive sampling method. The inclusion criteria included ischemic stroke patients who signed informed consent, were fully aware and oriented, and were able to complete the neurobehavior questionnaire. The exclusion criteria were patients who had symptoms and signs related to a previous neurobehavior disorder, aged more than 65 years old, and had severe comorbidities such as patients with a history of narcotics or alcohol abuse, functional psychiatric disorders, other structural diseases of the brain, acute coronary syndrome, diabetes mellitus, less than six years of formal education, and barrier of verbal communication. Written informed consent was signed, and the design was approved by the Research Ethics Committee, Dr. Moewardi General Hospital, with the ethical clearance number 208/II/HREC/2022.

The lipid profile, consisting of serum total cholesterol, LDL, HDL, and triglyceride levels, was checked on admission to the emergency unit. After fasting for 10-12 hours, blood samples were drawn and sent to the laboratory. Biochemical analysis was performed following centrifugation at 3500 rpm for 10 minutes. This study used Roche, Sigma, and Accurex reagents and kits. After patients had signed the informed consent form, the declining cognitive function was observed with the Montreal Cognitive Assessment Indonesian Version (MoCA-INA) score and Mini-Mental State Examination (MMSE) score in the wards.

The Kolmogorov-Smirnov test was used to assess the data distribution, while the Spearman correlation test was conducted to determine the correlation between lipid profile and cognitive function deterioration. Statistical analysis was continued with the ordinal regression test using backward elimination procedures to determine the odds ratio (OR) of cognitive function deterioration to lipid profile adjusted for covariates. The covariates assessed were gender, smoking history, obstructive sleep apnea (OSA), and body mass index (BMI). The statistical analysis was carried out using SPSS 25.0 for Windows.

RESULTS AND DISCUSSION

Data were collected from 68 ischemic stroke patients, and 52 met the inclusion and exclusion criteria. The majority of participants were male (n=29; 55.7%), and the mean age was 64.21 ± 9.78 years old, with a range of 35 to 72 years. The average BMI was 22.67 ± 5.82 kg/cm^2, and the participants consisted of 22 (43.2%) smokers and 30 (56.8%) non-smokers.

The results showed normal distribution after data were tested with the Kolmogorov-Smirnov test. The analysis was further conducted using the Spearman correlation test to analyze the lipid profile's correlation coefficient (r) with cognitive function deterioration. The results showed a moderate correlation with r=0.658 (p=0.04). Approximately 65.8% of cognitive function deterioration was affected by lipid profile, while the remaining 34.2% was influenced by other factors.

The data were further analyzed using ordinal logistic regression tests to determine the OR. The results showed that the OR of lipid profile including HDL, LDL, triglycerides, and total cholesterol were -1.54 (95% CI -0.98 - -3.46, p=0.002), 1.78 (95% CI 1.27-2.46, p=0.000), 2.48 (95% CI 1.39-4.92, p=0.04), and 1.54 (95% CI 1.04-2.48, p=0.003) respectively. This indicated that a declining level of HDL and increasing LDL, triglycerides, and total cholesterol were associated with the risk of cognitive function deterioration.

Factors contributing to cognitive function can be classified into unmodifiable, including sex, age, and genetics, as well as modifiable risk factors, namely diabetes, smoking, air pollution, depression, social isolation, and lack of physical activity (Baumgart et al., 2015; Rocca et al., 2014).

Based on the results, 29 (55.7%) male and 23 (44.3%) female participants were included. However, more severe cognitive function deterioration was found in female participants. In terms of gender, the results support a previous study, stating that women have a higher incidence rate of dementia compared to men at any given age. One hypothesis suggested that this phenomenon may be caused by hormonal and reproductive differences between the two genders (Ranson et al., 2021; Rocca et al., 2014).

A strong correlation was observed between low HDL levels as well as high triglyceride, LDL, and total cholesterol levels with the risk of impaired cognitive function. This supports a previous study conducted by Dimopoulos et al. (2015) stating that stroke patients with high triglyceride levels and low HDL cholesterol levels are more prone to memory impairment and depression. Other studies also showed that higher levels of total cholesterol and LDL in geriatrics correlated with faster cognitive decline (Dimopoulos et al., 2007; Ma et al., 2017).

According to previous studies, atherosclerosis is the main factor linking lipid profile with cognitive function. Dementia occurs earlier in patients with atherosclerosis through clinical and subclinical ischemic processes in the brain. In Alzheimer's disease, cerebrovascular and pathological changes often coexist and may act synergistically to accelerate cognitive decline. People with low serum HDL concentrations, as well as high levels of LDL, triglyceride, and total cholesterol, have an increased risk of stroke. Furthermore, the risk of developing Alzheimer's disease is elevated in stroke patients (Bandeali and Farmer, 2012; Feig et al., 2014).

Several mechanisms have been used to explain the correlation between low HDL cholesterol levels and the early onset of memory impairment. HDL can be found in plasma and cerebrospinal fluid (CSF). A primary component, apoE, facilitates cholesterol transport to the liver from peripheral tissues, including vascular and brain. HDL has a strong affinity to cytokine in the central nervous system and can penetrate through the blood-brain barrier. It also plays a significant role in the metabolism of Aβ protein, a component of amyloid plaques (Genin et al., 2011).

Antioxidant and anti-inflammatory properties of HDL may also contribute to neuronal degeneration in patients with dyslipidemia. It also accelerates synaptic maturation, enhances Aβ metabolism, maintains synaptic plasticity, and increases hippocampal volume. Lipid peroxidation, along with other oxidative stress substances, has been shown to increase the risk of developing Alzheimer's disease. Low HDL levels may impair cholesterol transport from brain cells, resulting in the accumulation and dysregulation of synaptic processes. This makes dyslipidemia a risk factor for neurodegenerative diseases (McGrowder et al., 2011; Panza et al., 2011).

Lifestyle modification can achieve neuroprotective effects when the lipid profile is well controlled. Several recommendations, including a balanced diet, physical activity, maintaining a normal BMI, and smoking cessation, can potentially increase HDL and decrease LDL, triglyceride, and total cholesterol levels. Furthermore, HDL can modify cholesterol ester transfer protein (CETP) activity, a neuroprotective protein associated with anti-dementia properties. It also reverses cholesterol transport mediated by the ABC transporter, thereby lowering cellular cholesterol and suppressing Aβ production. HDL inhibits Aβ oligomerization by directly binding to excess Aβ. Oligomerization is a major step in the production of neurotoxic compounds that cause memory impairment (Ashencho et al., 2017; Koch and Jense, 2016).

The study's limitation was its relatively small sample size. Further investigations with a bigger population and cohort design are needed to provide more comprehensive data. Besides, the MOCA-INA and MMSE questionnaires, which were self-administered, had some degree of bias, particularly recall bias. Certain variables related to dementia, including education, profession, and other socioeconomic status, were also not analyzed.

CONCLUSION

The results showed a positive correlation with moderate strength between lipid profile and cognitive function deterioration in ischemic stroke patients.

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